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## LECTURE 06-Memory

*Remembering, Forgetting, and How Memory Works*

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### 1. What Is Memory?

#### Definition

Memory is the mental system that **encodes**, **stores**, and **retrieves** information. It allows us to learn from experiences, recognize familiar people or objects, solve problems, and maintain our sense of identity.

#### Explanation

Memory is not like a video camera. It is an active process shaped by:

- **Bottom-up inputs** — sensory data (what we see/hear)
- **Top-down influences** — prior knowledge, expectations, culture, emotions

#### Mini Diagram

Experience → Encoding → Storage → Retrieval → Behavior/Decision

#### Example

You recognize your friend's voice (stored auditory memory) when they call you.

#### Case Example

A student remembers exam content better when they connect new concepts with old knowledge — showing top-down influence.

#### Memory Trick

Think **ESR** — Encoding, Storage, Retrieval (like “EdiSon’s Record”).

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### 2. Bottom-Up vs Top-Down Influences on Memory

#### Bottom-Up Processing

Information flows from **sensation** → **perception** → **memory**.

**Example:**

You hear a loud bang. You remember the sound because it was surprising.

**Top-Down Processing**

Your memory is shaped by your **existing knowledge, culture, and expectations**.

**Example:**

If you believe cats are playful, you may remember a cat video as funnier than it was.

**Cultural Insight**

People from collectivist cultures tend to recall **social interactions** better, while individualistic cultures recall **personal achievements** more strongly.

**Mini Diagram**

BOTTOM-UP: Sensory → Meaning

TOP-DOWN: Meaning → Interpretation of Sensory

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**3. The Three Stages of Memory**

**3.1 Encoding**

**Definition**

The process of transforming sensory input into a meaningful memory trace.

**Types of Encoding**

| Type            | Description   | Example                                 |
|-----------------|---------------|-----------------------------------------|
| <b>Visual</b>   | Images        | Remembering a logo                      |
| <b>Auditory</b> | Sounds        | Remembering a melody                    |
| <b>Semantic</b> | Meaning-based | Remembering “democracy = people’s rule” |

**Psychological Insight**

Deep, meaningful processing activates more brain areas → stronger memories.

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**3.2 Storage**

**Definition**

Maintaining information over time.

**Example**

Your ability to remember multiplication tables years later.

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**3.3 Retrieval****Definition**

Accessing stored information when needed.

**Example**

You recall your ATM PIN when withdrawing money.

**Application**

Self-testing strengthens retrieval pathways.

**Mini Diagram**

Encoding → (Storage) → Retrieval → Performance

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**4. Three-Stage Model of Memory**

Sensory Memory → Short-Term Memory → Long-Term Memory  
(attention)      (rehearsal)

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**5. Sensory Memory****Definition**

Brief storage of raw sensory information (visual, auditory, smell, taste, touch).

**Types**

| Type                     | Duration    | Example                         |
|--------------------------|-------------|---------------------------------|
| <b>Iconic</b> (visual)   | 0.3–0.5 sec | Afterimage of a sparkler        |
| <b>Echoic</b> (auditory) | 2–4 sec     | “What did you say? ... oh yes.” |
| <b>Haptic</b> (touch)    | 1–2 sec     | Feeling of a tap                |

**Key Properties**

- Very accurate but fades quickly
- Helps us perceive continuity (e.g., movies appear smooth)

### Example Activity

“What letters did you just see?” — illustrates iconic memory.

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## 6. Short-Term Memory (STM)

### Definition

A temporary workspace where information is held for about **15–30 seconds**.

**Capacity: “ $7 \pm 2$  chunks” (Miller, 1956)**

### Chunking

Meaningful grouping of information to improve capacity.

### Example:

“17762024” → “1776 | 2024” (2 chunks instead of 8 digits)

### Digit Span Test (from lecture)

Shows increasing chunk capacity from 4 to 11 digits.

### How Prior Knowledge Helps

- Helps form meaningful chunks
- BUT may distort memory (top-down bias)

### Mini Diagram

INFO → STM (limited) → rehearsal → LTM

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## 7. Long-Term Memory (LTM)

### Definition

Permanent, nearly limitless storage of knowledge, skills, and experiences.

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### 7.1 Implicit (Non-declarative) Memory

## Definition

Memory retrieved *without conscious awareness*.

## Types & Examples

| Type                  | Example                                    |
|-----------------------|--------------------------------------------|
| Procedural            | Riding a bike                              |
| Priming               | Seeing “doctor” makes you think of “nurse” |
| Conditioned responses | Fear of dogs after a bite                  |

## Mini Case

A car driver automatically uses brake+clutch timing — procedural memory.

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## 7.2 Explicit (Declarative) Memory

### Definition

Memory retrieved *with conscious effort*.

### Types

| Type     | Description          | Example                   |
|----------|----------------------|---------------------------|
| Episodic | personal experiences | Your 10th birthday party  |
| Semantic | facts, knowledge     | Capital of France = Paris |

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### Implicit vs Explicit — Comparison Table

| Feature     | Implicit                  | Explicit      |
|-------------|---------------------------|---------------|
| Awareness   | Unconscious               | Conscious     |
| Retrieval   | Automatic                 | Effortful     |
| Examples    | Skills, priming           | Facts, events |
| Brain Areas | Basal ganglia, cerebellum | Hippocampus   |

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## 8. Effects That Influence Memory Encoding

### 8.1 Primacy and Recency Effect

#### Explanation

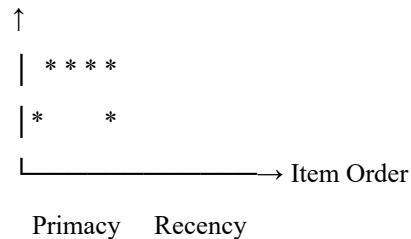
In a list:

- **Primacy** → better recall of early items (LTM)

- **Recency** → better recall of later items (STM)

### Diagram

Recall Accuracy



### Example

You remember the first and last items in a shopping list better than the middle.

## 8.2 Levels of Processing (Craik & Lockhart, 1972)

### Shallow Encoding

Focus on appearance (capital letters).

→ Poor memory.

### Deep Encoding

Focus on meaning.

→ Strong memory.

### Experiment Example

Participants recall more words when asked:

“Does this word relate to something alive?” than

“Is this word written in capital letters?”

### Table

| Encoding Type | Focus        | Outcome       |
|---------------|--------------|---------------|
| Shallow       | Looks, sound | Weak memory   |
| Deep          | Meaning      | Strong memory |

## 8.3 Organization Matters

Semantically organized lists (e.g., “Career”, “Home”, “Food”) lead to better recall than random lists.

### Example:

Grouping “LAWYER, PHYSICIAN, TEACHER” under “PROFESSIONAL” improves memory

### Mini Diagram

Random → weak memory

Organized → strong memory

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## 8.4 Context-Dependent Memory

You recall better when in the same environment in which you learned.

### Example:

Studying in your room → better recall during the exam if mental context matches.

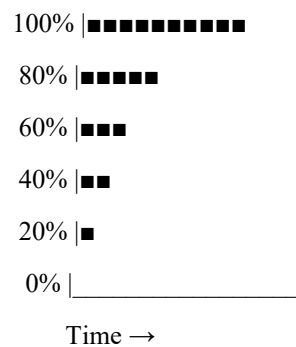
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## 9. Forgetting

### 9.1 Ebbinghaus Forgetting Curve

Memory loss is rapid at first, then plateaus.

#### Diagram (Text-based)



### Example

You forget most of a lecture within days unless reviewed.

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## 9.2 Interference Theory

### Proactive Interference

Old information blocks new information.

### Example:

You keep typing your old password when trying to memorize a new one.

### Retroactive Interference

New information disrupts old information.

### Example:

Learning a new phone number makes you forget your old phone number.

#### **Table**

| <b>Type</b> | <b>Old vs New</b>         | <b>Example</b>              |
|-------------|---------------------------|-----------------------------|
| Proactive   | Old → interferes with new | Old password blocks new     |
| Retroactive | New → interferes with old | New student names erase old |

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## **10. Flashbulb Memories**

### **Definition**

Extremely vivid, emotionally charged memories of dramatic events.

### **Features**

- Caused by surprise + emotional arousal
- High confidence, but not always accurate
- Often distorted over time

### **Examples**

- Remembering where you were during a major tragedy
- Recalling news of a national event

### **Cultural Insight**

Communities often share “collective flashbulb memories,” strengthening group identity.

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## **11. Improving Memory (Scientifically Proven Strategies)**

### **1. Repeated, spaced study**

→ Better than last-minute cramming.

### **2. Elaborative rehearsal**

Connect with meaning, stories, personal examples.

### **3. Self-reference effect**

Link material to your life.

### **Example:**



“Proactive interference is like when I forget my new Wi-Fi password.”

#### 4. Retrieval practice

Testing yourself strengthens pathways.

#### 5. Reduce interference

Study in a quiet room.

#### 6. Sleep

Consolidates memories into LTM.

#### 7. Exercise

Boosts blood flow + neurogenesis.

#### 8. Organization

Use diagrams, tables, categories.

#### 9. Brain-training apps

Mildly helpful for attention and working memory.

#### 10. Healthy lifestyle

Diet, socializing, staying mentally active.

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#### 12. Final Summary Table — MEMORY (Cheat Sheet)

| Topic                  | Core Idea                     | Key Example           |
|------------------------|-------------------------------|-----------------------|
| <b>Encoding</b>        | Transforming info into memory | Semantic > visual     |
| <b>Storage</b>         | Keeping info over time        | STM 15–30 sec         |
| <b>Retrieval</b>       | Finding stored info           | Self-testing          |
| <b>Sensory Memory</b>  | Brief sensory storage         | Iconic = 0.5 sec      |
| <b>STM</b>             | $7 \pm 2$ chunks              | Phone numbers         |
| <b>LTM</b>             | Permanent store               | Facts, skills         |
| <b>Implicit</b>        | Unconscious memory            | Riding a bike         |
| <b>Explicit</b>        | Conscious recall              | Birthdays             |
| <b>Primacy/Recency</b> | Serial position effects       | Remember first & last |

| Topic            | Core Idea                         | Key Example            |
|------------------|-----------------------------------|------------------------|
| Deep vs Shallow  | Meaning > appearance              | “Living?” > “CAPITAL?” |
| Interference     | Old ↔ New conflict                | Password confusion     |
| Flashbulb        | Emotional memories                | Tragic events          |
| Improving Memory | Spaced repetition, sleep, meaning | Study strategies       |

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